

DRAWVLA-VERIFY: VALIDATING SKETCHED ROBOT INSTRUCTIONS BEFORE EXECUTION

Team Member 1, Team Member 2 & Team Member 3*

School of Computing and Information Systems

Singapore Management University

{member1, member2, member3}@smu.edu.sg

ABSTRACT

Sketches let a person tell a robot *which* object to manipulate (a circle) and *where* it should go (an arrow) without naming anything, but existing sketch-conditioned robot systems assume the drawing is correct. We propose **sketch-prompt validation**—deciding *before* any action whether a drawn instruction is correct, consistent across circle, arrow, and caption, and sufficient for a unique action, and saying why not—as an explicit, benchmarked task. A pilot simulator-grounded generator over LIBERO yields 37,970 labelled prompts; frontier vision-language models (VLMs) fail it zero-shot (best 0.79 accuracy), a 500M LoRA-tuned validator reaches 0.90, yet a geometry-only classifier reaches 0.93, exposing shortcuts. We propose to (H1) confirm learnability against strong zero-shot and fine-tuned baselines, (H2) remove shortcuts with a distribution-matched, adversarially filtered generator audited by scene-blind ceilings, (H3) *generate* a per-predicate verdict and rejection reason behind a conformal false-accept guarantee, and (H4) test transfer to real human sketches.

1 PROBLEM AND MOTIVATION

Language is ambiguous in cluttered scenes with duplicate objects and goal images are over-specified (Sundaresan et al., 2024); a circle-and-arrow sketch with a name-free caption (“move this there”) is a persistent deictic act (Saupé & Mutlu, 2014; Kranstedt et al., 2006). Recent systems condition robot policies on such sketches (Sundaresan et al., 2024; Gu et al., 2024; Li et al., 2025a;b) and all silently assume the drawing is *valid*. Yet circles miss or enclose a distractor, arrows point to the wrong container or detach from the circle, captions contradict sketches, and channels that look fine alone can be jointly inconsistent; executing such a sketch commits the robot to the wrong action.

Problem (Track B). *Can a model decide, before any action, whether a drawn instruction is safe to execute—and say why not?* We formulate **sketch-prompt validation** as a decision over image, circle, arrow, and caption requiring per-channel correctness, cross-channel consistency, and sufficiency (Section 3.1), plus a generated reason; no existing task captures it (Section 2). We address it with generative VLMs as zero-shot judges and as LoRA-tuned validators.

Pilot evidence and gap. A pilot by team members (DrawVLA-Verify pilot, 2026) built DrawVLA-Verify, a simulator-grounded generator over LIBERO (Liu et al., 2023a) producing 37,970 labelled prompts across 40 tasks with no manual annotation. Frontier VLMs fail zero-shot (best 0.79 accuracy, 0.43 MCC; majority class 0.74) while SmolVLM-500M (Marafioti et al., 2025) with LoRA (Hu et al., 2022) reaches 0.90 (Table 3, Appendix B); but a geometry-only classifier reaches 0.93, i.e., the generator leaks shortcuts (Geirhos et al., 2020), and the pilot was single-seed with one zero-shot prompt and no ablations, human sketches, or explanations. This project delivers those deltas.

Hypotheses. H1 (learnability): a ≤ 1 B LoRA-tuned VLM beats the best zero-shot VLM under its best prompt by ≥ 10 accuracy points and is within 1 balanced-accuracy point of, or exceeds, a fine-tuned discriminative model of similar size, over 3 seeds. **H2 (shortcut removal):** on the hardened tier the multimodal validator keeps ≥ 0.80 balanced accuracy and exceeds both scene-blind ceilings

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(geometry-only and sketch-on-blank-canvas, measured on the untouched audit split) by ≥ 15 points. **H3 (structured supervision):** supervising the five validity predicates as auxiliary generative targets improves the directional and grounding slices (the pilot’s weakest) by ≥ 3 points over binary-only training, and generated reasons name the ground-truth violated predicate with $\geq 80\%$ faithfulness. **H4 (sim-to-human transfer):** validators trained with guarded imprecision augmentation lose ≤ 10 balanced-accuracy points on real human sketches; removing it widens the gap by ≥ 5 points. **Gate:** with $\hat{\tau}$ from Eq. (2), empirical test FAR is ≤ 0.05 in every seed on v1 and v2; on human sketches we report the realised FAR and read violations as distribution shift.

2 RELATED WORK

Sketch- and visual-prompt-based instruction. RT-Sketch and RT-Trajectory condition policies on goal or trajectory sketches (Sundaresan et al., 2024; Gu et al., 2024), RoVI decodes an arrow/circle language with a VLM (Li et al., 2025b), CrayonRobo is *tolerant* of noisy prompts but does not classify them (Li et al., 2025a), and Action-Sketcher and AnyUser generate or translate sketches (Tan et al., 2026; Yang et al., 2026). Visual-prompting methods (PIVOT, Set-of-Mark, MOKA, RoboPoint) (Nasiriany et al., 2024; Yang et al., 2023; Liu et al., 2024; Yuan et al., 2024) annotate images so a VLM can ground spatial decisions, and ViP-LLaVA reads arbitrary overlaid marks (Cai et al., 2024). All assume the prompt is valid; *we extend* this line by making validity the primary decision, keeping their circle+arrow+caption format so a validator can sit in front of any of them.

Verification and safety gating. KnowNo lets a planner ask for help via conformal prediction (Ren et al., 2023); VerifyLLM and VeriGraph verify plans against temporal logic or scene graphs (Sautenkov et al., 2025; Ekpo et al., 2024); DoReMi and REFLECT detect or explain execution failures (Guo et al., 2024; Liu et al., 2023b). These verify *plans* or *executions*; *we contrast* by verifying the instruction itself, an earlier gate, with a conformal threshold (Angelopoulos et al., 2024; Geifman & El-Yaniv, 2017).

Shortcuts. A geometry-only classifier is a hypothesis-only baseline in the NLI sense (Gururangan et al., 2018); AFLite (Le Bras et al., 2020) and dataset cartography (Swayamdipta et al., 2020) remove or expose such artefacts (Geirhos et al., 2020) in fixed datasets, whereas *we control the generator* and can both re-sample and filter.

3 TASK FORMULATION AND METHOD SKETCH

3.1 FORMULATION

A sketch prompt is $P = (I, C, A, L)$: RGB observation I , circle C (centre, radius) denoting *which* object, arrow A (tail, head) denoting *where* it goes, and a name-free caption L . From simulator ground truth we define predicates v_C (the circle uniquely and correctly selects the target), v_A (correct bearing and magnitude, anchored on the circled object, ending in the intended region), v_L (caption verb and implicit referents match the intended action), cross-modal consistency v_X (all channels describe the same object, destination, and predicate), and sufficiency v_S (the prompt determines a unique executable action). A prompt is valid iff $V(P) = v_C(P) \wedge v_A(P) \wedge v_L(P) \wedge v_X(P) \wedge v_S(P)$, and the task is to learn $f_\theta(P) \in \{\text{ACCEPT}, \text{REJECT}\}$ predicting $V(P)$, with a reason on REJECT. A prompt can satisfy every per-channel predicate yet be invalid because $v_X = 0$ or $v_S = 0$, so validation cannot be reduced to independent per-channel checks.

3.2 SIMULATOR-GROUNDED DATA GENERATION AND SHORTCUT HARDENING

Pilot pipeline (Figure 1, top). From LIBERO’s object poses, region geometry, and gripper trajectories we extract the correct circle target and arrow endpoints for every frame with no human labelling. Valid prompts get *guarded* human-imprecision augmentation (jitter, non-circular strokes, hand-like shafts): an in-point guard keeps the target inside the circle, an out-point guard keeps distractors outside. Negatives apply one controlled corruption at a time from four families (14 of 18 specified modes implemented in the pilot; Appendix A) plus five grounding configurations that stress ambiguity and sufficiency, so every negative has a known mode, severity, and violated predicate. Captions are name-free so the sketch carries the instruction.

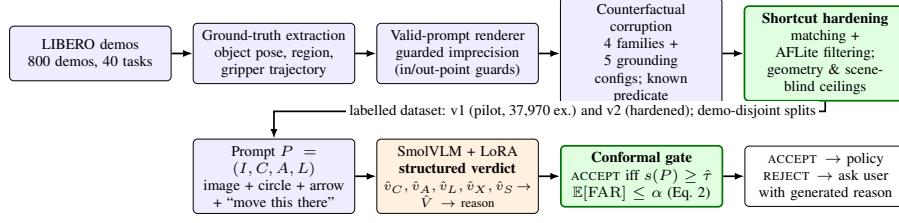


Figure 1: Method sketch. *Top*: simulator-grounded data generation (green = new). *Bottom*: the validator generates a structured verdict and a reason; a conformal threshold yields an execute/ask decision.

New: shortcut hardening (H2). Let $g(P)$ be the raw geometry vector (circle centre and radius, arrow endpoints, length, bearing). (i) *Distribution matching*: on the valid side of a guard, negatives draw the same magnitudes as valid imprecision; corruptions that must cross a guard (near-miss, wrong magnitude) draw severities from the mirrored distribution just beyond it, so class-conditional marginals of g overlap as far as labels allow. (ii) *Adversarial filtering* (Le Bras et al., 2020) on medium/hard modes only (easy modes such as whole-frame circles are geometrically invalid by definition): we repeatedly train geometry-only gradient-boosted classifiers on random folds, score each example by how often held-out classifiers predict it correctly with confidence > 0.75 , and remove the most predictable until geometry-only balanced accuracy on an *untouched* audit split (also re-scored with MLP and k -NN adversaries) is ≤ 0.60 or retention falls below 60%, reporting per-mode retention. Because pixel artefacts can leak even when coordinates do not, we also report a *scene-blind* ceiling: the validator trained and tested on sketches on a blank canvas.

3.3 STRUCTURED VALIDATOR AND CONFORMAL INFERENCE

Model. A compact VLM (SmoIVLM-256M/500M) with LoRA takes the image with the rendered sketch, the caption, and a fixed prompt and generates a *structured verdict*: five predicate questions q_k , $k \in \{C, A, L, X, S\}$, answered sequentially with earlier answers in context (teacher-forced in training), then the ACCEPT/REJECT question q_V , then, on REJECT, a one-sentence reason. Predicate labels are free (each negative violates a known subset; positives satisfy all) and close to a family label; we also report the derived decision $\hat{V}' = \bigwedge_k \hat{v}_k$ and the verdict–predicate inconsistency rate.

Objective. Each yes/no answer is scored by forced choice over its two answer tokens, $p_\theta(\text{yes} \mid P, q_k) = \sigma(z_{\text{yes}} - z_{\text{no}})$ with z the next-token logits; the reason uses token-level cross-entropy ℓ_R , on REJECT examples only, against one of ≥ 3 paraphrased template reasons per failure mode (one paraphrase per mode held out from training). We minimise

$$\mathcal{L}(\theta) = \ell_V(\theta) + \lambda \sum_{k \in \{C, A, L, X, S\}} \ell_k(\theta) + \mu \ell_R(\theta), \quad \ell_k(\theta) = -\log p_\theta(y_k \mid P, q_k), \quad (1)$$

with λ, μ tuned on validation; $\lambda = \mu = 0$ recovers the pilot’s binary-only validator (ablation A1).

Inference with a finite-sample false-accept guarantee. Let $s(P) = p_\theta(\text{ACCEPT} \mid P, q_V)$; we ACCEPT iff $s(P) \geq \hat{\tau}$. False accepts (executing an invalid prompt) are the safety-critical error, so we bound the false-accept rate (FAR) by conformal risk control (Angelopoulos et al., 2024): on the n *invalid* validation prompts let $\ell_i(\tau) = \mathbb{1}[s(P_i) \geq \tau]$ (non-increasing in τ , bounded by 1) and

$$\hat{\tau} = \inf \left\{ \tau : \frac{1}{n+1} \sum_{i=1}^n \ell_i(\tau) + \frac{1}{n+1} \leq \alpha \right\}. \quad (2)$$

If invalid validation and test prompts are exchangeable, $\mathbb{E}[\text{FAR}(\hat{\tau})] \leq \alpha$ ($\alpha = 0.05$); we report the induced false-reject rate and empirical coverage across seeds, tiers, and the human set (where exchangeability is *not* guaranteed), and rejected prompts return the generated reason as a clarification request.

4 DATASETS, METRICS, BASELINES, AND EXPERIMENTAL PLAN

Datasets and licences. (1) *LIBERO* (Liu et al., 2023a) (MIT; scene assets under robosuite/MuJoCo terms): 800 demonstrations, 40 tasks. (2) *DrawVLA-Verify v1* (pilot): 37,970 examples—30,302

train / 3,702 val / 3,966 test, demo-disjoint; 9,098 positives, 28,872 negatives (Appendix A)—each with image, coordinates, caption, label, and mode. (3) *v2 (hardened)*: same splits after matching and filtering ($\geq 60\%$ retention expected) plus the untouched audit split. (4) *Human sketch test set* (evaluation only; SMU IRB exemption sought first): ≈ 10 adult volunteers each draw circle+arrow prompts (mouse or tablet) on 30 demo-disjoint frames, 15 *unprompted* (draw quickly) and 15 *deliberately wrong*, giving ≈ 300 sketches (≈ 150 per condition; metrics balanced); no captions, so v_L/v_X are out of scope. Labels come from the guards; two members annotate blind to them and disagreement is reported as a finding. Data and adapters: CC BY 4.0; code: MIT.

Metrics. Accuracy, balanced accuracy, macro-F1, MCC (74% negative test set); FAR and FRR at $\hat{\tau}$ and FAR at FRR = 10%; per-family and per-mode slice accuracy; geometry-only and scene-blind ceilings per tier; the transfer gap Δ_{human} (balanced accuracy); reason faithfulness on true rejects against (a) the ground-truth predicate and (b) the model’s own $\hat{v}$ (self-consistency), by string match and an open LLM judge (Qwen3-8B) (false-reject reasons reported qualitatively); and a 5-point human usefulness rating of reasons on 50 rejected items.

Baselines. (i) Majority class. (ii) Zero-shot open VLMs Qwen3-VL-2B/4B/8B (Qwen Team, 2025) (Apache-2.0) and (iii) closed APIs (Gemini 3.7 Flash, Claude Haiku 4.5/Sonnet 4.6) with pinned versions, cached outputs, costs and rate limits reported, and no conclusion resting solely on them; for each model the best of direct, chain-of-thought, 4-shot, and coordinates-as-text prompts is chosen on a 500-example stratified validation subset and run once on test. (iv) Geometry-only gradient boosting (Ke et al., 2017) and the scene-blind validator as ceilings. (v) A fine-tuned discriminative model of similar size (ConvNeXt-T with late fusion of $g(P)$ and a caption embedding). (vi) The pilot’s binary-only SmolVLM+LoRA. *Ours*: structured SmolVLM+LoRA with the conformal gate.

Ablations (≥ 3), sensitivity, statistics. A1: structured (+reason) vs. binary-only ($\lambda = \mu = 0$; H3). A2: train on v2 vs. v1, test on both tiers (H2). A3: modality (no caption; coordinates-as-text without pixels; scene-blind). A4: no guarded-imprecision augmentation (H4). A5: prompt-only vs. LoRA. A6: backbone 256M/500M/2.2B. Sensitivity: training-set size (10/25/50/100%), LoRA rank (4/16/64), τ sweep (FAR–FRR curve), corruption-severity sweeps (near-miss, wrong magnitude). All trained models use 3 seeds (mean $\pm$ std); McNemar’s test (Dietterich, 1998) on seed-matched predictions (p -values combined across seeds), paired-bootstrap 95% CIs, and minimum detectable effects per slice (directional+grounding: 1,613 test items).

5 RISKS, RESOURCES, AND TEAM PLAN

Risks and mitigations. **Hardening over-removes or still leaks:** per-mode retention curves, removal capped at 40%, both tiers released. **Small human set; guard label noise:** evaluation-only probe with bootstrap CIs; blind audit of 200 positives and 200 negatives, flagging modes with agreement < 0.8 . **API drift, cost, compute:** pinned versions, cached pilot outputs, Qwen3-VL primary, ablations on the 256M model. **Schedule:** go/no-go 25 Oct—if the v2 ceiling is still ≥ 0.75 at 60% retention (H2 is tested at whatever ceiling is reached) we report v1 plus the best partial tier and shift effort to H3/H4. **Over-claiming safety:** the gate reduces, not eliminates, unsafe executions; no closed-loop control is evaluated.

Compute. One NVIDIA A100-40GB or RTX 4090-24GB: ≈ 3 GPU-h per SmolVLM-500M LoRA run (3 epochs, batch 64, bf16, peak memory < 20 GB), ≈ 1.5 h per 256M, ≈ 8 h per 2.2B; ≈ 45 runs ≈ 160 GPU-h plus ≈ 10 GPU-h of inference. APIs ($\approx 4k$ images $\times$ 3 models $\times$ 4 prompts; pilot outputs reused) $\approx$ USD 100–150.

Roles. *Member 1 (data):* generator v2, hardening, audit, release, prompt sweep, ConvNeXt baseline. *Member 2 (modelling):* structured objective, reasons, conformal gate, LoRA training over seeds and scales, ablations. *Member 3 (evaluation):* API baselines, statistics, human study, ethics. **Timeline (2026).** 1–11 Oct: IRB, recruitment. 12–25 Oct: generator v2, 3-seed re-run with stronger baselines (H1), validator and gate, A1–A3; go/no-go. 26 Oct–8 Nov: human study (H4), A4–A6, sensitivity, statistics, artifacts. 9–12 Nov: final presentation ($\geq 90\%$ of results); 13–27 Nov: final report and artifacts.

ETHICS STATEMENT

All training data are simulator renders of tabletop scenes with no people or personal information. The human sketch study involves adult volunteers who give informed consent; only drawings and anonymous ratings are stored. The validator is a trust gate for robot instructions: a false accept could cause an incorrect physical action, so we report false-accept rates with confidence intervals and recommend deployment with human confirmation. We comply with the MIT licence of LIBERO, the Apache-2.0 licences of SmolVLM and Qwen3-VL, and the terms of service of the closed APIs, and release our data under CC BY 4.0 and code under MIT.

REPRODUCIBILITY STATEMENT

We will release the generator, training and evaluation code, `configs/*.yaml`, seeds, an `environment.yml`, a one-click `scripts/run.sh`, cached API outputs, LoRA adapters (<2 GB), and both dataset tiers with checksums, together with GPU model, GPU-hours, and peak memory (Section 5). Pilot training details and the zero-shot prompt are in Appendix B.

AI USE STATEMENT

[To be completed by the team before submission, in accordance with the CS707 academic-integrity policy.]

REFERENCES

- Anastasios N. Angelopoulos, Stephen Bates, Adam Fisch, Lihua Lei, and Tal Schuster. Conformal risk control. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2208.02814.
- Mu Cai, Haotian Liu, Siva Karthik Mustikovela, Gregory P. Meyer, Yuning Chai, Dennis Park, and Yong Jae Lee. ViP-LLaVA: Making large multimodal models understand arbitrary visual prompts. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. arXiv:2312.00784.
- Thomas G. Dietterich. Approximate statistical tests for comparing supervised classification learning algorithms. *Neural Computation*, 10(7):1895–1923, 1998.
- DrawVLA-Verify pilot. DrawVLA-Verify: Learning to validate circle-and-arrow robot instructions from simulator ground truth. Manuscript (pilot study); venue status to be stated, 2026.
- Daniel Ekpo, Mara Levy, Saksham Suri, De-An Huang, Dinesh Manocha, et al. VeriGraph: Scene graphs for execution verifiable robot planning. *arXiv preprint arXiv:2411.10446*, 2024.
- Yonatan Geifman and Ran El-Yaniv. Selective classification for deep neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30, 2017.
- Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A. Wichmann. Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2(11):665–673, 2020.
- Jiayuan Gu, Sean Kirmani, Paul Wohlhart, Yao Lu, Montserrat Gonzalez Arenas, Kanishka Rao, Wenhao Yu, Chuyuan Fu, Keerthana Gopalakrishnan, Zhuo Xu, Priya Sundareshan, Peng Xu, Hao Su, Karol Hausman, Chelsea Finn, Quan Vuong, and Ted Xiao. RT-Trajectory: Robotic task generalization via hindsight trajectory sketches. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2311.01977.
- Yanjiang Guo, Yen-Jen Wang, Lihan Zha, Zheyuan Jiang, and Jianyu Chen. DoReMi: Grounding language model by detecting and recovering from plan-execution misalignment. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024. arXiv:2307.00329.

- Suchin Gururangan, Swabha Swayamdipta, Omer Levy, Roy Schwartz, Samuel R. Bowman, and Noah A. Smith. Annotation artifacts in natural language inference data. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, pp. 107–112, 2018.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*, 2022. arXiv:2106.09685.
- Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30, 2017.
- Alfred Kranstedt, Stefan Kopp, and Ipke Wachsmuth. Deixis in multimodal human-computer interaction: An interdisciplinary approach. In *Gesture-Based Communication in Human-Computer Interaction*, volume 3881 of *Lecture Notes in Computer Science*, pp. 112–123. Springer, 2006.
- Ronan Le Bras, Swabha Swayamdipta, Chandra Bhagavatula, Rowan Zellers, Matthew E. Peters, Ashish Sabharwal, and Yejin Choi. Adversarial filters of dataset biases. In *Proceedings of the 37th International Conference on Machine Learning (ICML)*, volume 119 of *Proceedings of Machine Learning Research*, pp. 1078–1088, 2020.
- Xiaoqi Li, Jingyun Xu, Mingxu Zhang, Jiaming Liu, Yan Shen, Iaroslav Ponomarenko, Jiahui Xu, Liang Heng, Siyuan Huang, Shanghang Zhang, and Hao Dong. CrayonRobo: Object-centric prompt-driven vision-language-action model for robotic manipulation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025a. arXiv:2505.02166.
- Yanbang Li, Ziyang Gong, Haoyang Li, Xiaoqi Huang, Haolan Kang, Guangping Bai, and Xi-anzheng Ma. Robotic visual instruction. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 12155–12165, 2025b. arXiv:2505.00693.
- Bo Liu, Yifeng Zhu, Chongkai Gao, Yihao Feng, Qiang Liu, Yuke Zhu, and Peter Stone. LIBERO: Benchmarking knowledge transfer for lifelong robot learning. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, 2023a. arXiv:2306.03310.
- Fangchen Liu, Kuan Fang, Pieter Abbeel, and Sergey Levine. MOKA: Open-world robotic manipulation through mark-based visual prompting. In *Robotics: Science and Systems (RSS)*, 2024. arXiv:2403.03174.
- Zeyi Liu, Arpit Bahety, and Shuran Song. REFLECT: Summarizing robot experiences for failure explanation and correction. In *Proceedings of the 7th Conference on Robot Learning (CoRL)*, volume 229 of *Proceedings of Machine Learning Research*, 2023b. arXiv:2306.15724.
- Andrés Marafioti, Orr Zohar, Miquel Farré, Merve Noyan, Elie Bakouch, Pedro Cuenca, Cyril Zakkka, Loubna Ben Allal, Anton Lozhkov, Nouamane Tazi, Vaibhav Srivastav, Joshua Lochner, Hugo Larcher, Mathieu Morlon, Lewis Tunstall, Leandro von Werra, and Thomas Wolf. SmolVLM: Redefining small and efficient multimodal models. *arXiv preprint arXiv:2504.05299*, 2025.
- Soroush Nasiriany, Fei Xia, Wenhao Yu, Ted Xiao, Jacky Liang, Ishita Dasgupta, Annie Xie, Danny Driess, Ayzaan Wahid, Zhuo Xu, Quan Vuong, Tingnan Zhang, Tsang-Wei Edward Lee, Kuang-Huei Lee, Peng Xu, Sean Kirmani, Yuke Zhu, Andy Zeng, Karol Hausman, Nicolas Heess, Chelsea Finn, Sergey Levine, and Brian Ichter. PIVOT: Iterative visual prompting elicits actionable knowledge for VLMs. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *Proceedings of Machine Learning Research*, pp. 37321–37341, 2024.
- Qwen Team. Qwen3-VL technical report. <https://github.com/QwenLM/Qwen3-VL>, 2025.

- Allen Z. Ren, Anushri Dixit, Alexandra Bodrova, Sumeet Singh, Stephen Tu, Noah Brown, Peng Xu, Leila Takayama, Fei Xia, Jake Varley, Zhenjia Xu, Dorsa Sadigh, Andy Zeng, and Anirudha Majumdar. Robots that ask for help: Uncertainty alignment for large language model planners. In *Proceedings of the 7th Conference on Robot Learning (CoRL)*, volume 229 of *Proceedings of Machine Learning Research*, 2023. arXiv:2307.01928.
- Allison Saupé and Bilge Mutlu. Robot deictics: How gesture and context shape referential communication. In *Proceedings of the 2014 ACM/IEEE International Conference on Human-Robot Interaction (HRI)*, pp. 342–349, 2014.
- Oleg Sautenkov et al. VerifyLLM: LLM-based pre-execution task plan verification for robots. *arXiv preprint arXiv:2507.05118*, 2025.
- Priya Sundareshan, Quan Vuong, Jiayuan Gu, Peng Xu, Ted Xiao, Sean Kirmani, Tianhe Yu, Michael Stark, Ajinkya Jain, Karol Hausman, Dorsa Sadigh, Jeannette Bohg, and Stefan Schaal. RT-Sketch: Goal-conditioned imitation learning from hand-drawn sketches. In *Proceedings of the 8th Conference on Robot Learning (CoRL)*, volume 270 of *Proceedings of Machine Learning Research*, 2024. arXiv:2403.02709.
- Swabha Swayamdipta, Roy Schwartz, Nicholas Lourie, Yizhong Wang, Hannaneh Hajishirzi, Noah A. Smith, and Yejin Choi. Dataset cartography: Mapping and diagnosing datasets with training dynamics. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 9275–9293, 2020.
- Hao Tan, Peter Co, Yuchen Xu, Shuai Rong, Yatao Ji, Cheng Chi, Xu Chen, Qiang Zhang, Zhen Zhao, Peng Wang, Zhe Wang, and Shanghang Zhang. Action-sketcher: From reasoning to action via visual sketches for long-horizon robotic manipulation. *arXiv preprint arXiv:2601.01618*, 2026.
- Jianwei Yang, Hao Zhang, Feng Li, Xueyan Zou, Chunyuan Li, and Jianfeng Gao. Set-of-mark prompting unleashes extraordinary visual grounding in GPT-4V. *arXiv preprint arXiv:2310.11441*, 2023.
- Shuo Yang, Hao Tan, Kai Yang, Wei Yang, and Sheng Yang. AnyUser: Translating sketched user intent into domestic robots. *arXiv preprint arXiv:2604.04811*, 2026. Accepted to IEEE Transactions on Robotics.
- Wentao Yuan, Jiafei Duan, Valts Blukis, Wilbert Pumacay, Ranjay Krishna, Adithyavairavan Murali, Arsalan Mousavian, and Dieter Fox. RoboPoint: A vision-language model for spatial affordance prediction in robotics. In *Proceedings of the 8th Conference on Robot Learning (CoRL)*, volume 270 of *Proceedings of Machine Learning Research*, pp. 4005–4020, 2024. arXiv:2406.10721.

A FAILURE TAXONOMY AND DATASET COMPOSITION

Table 1: Failure taxonomy. Difficulty: E(asy)/M(edium)/H(ard). Each mode has a controlled generation mechanism and maps to the predicate(s) it violates. Of the 18 corruption modes listed, 14 are implemented in the pilot release; the remaining four are scheduled for v2.

Family	Mode	Description and generation mechanism	Diff.
Referential (circle / “which”; v_C)	Wrong-category object	Circle a clearly different object (e.g., plate, stove) than the target.	E
	Near-miss circle	Offset the centre so the target falls outside or only grazes the circle; severity varies continuously with offset.	H
	Empty-region circle	Circle bare table or background containing no object.	E
	Multi-object enclosure	Inflate the radius to enclose the target <i>and</i> a distractor, making the referent unresolvable (tests v_S).	H
	Degenerate size	Dot-sized or whole-frame circle that selects nothing meaningful.	E
Directional (arrow / “where”; v_A)	Wrong destination	Retarget the arrow head to a different container or region.	H
	Wrong magnitude	Correct bearing but arrow far too short or overshooting.	H
	Wrong bearing	Correct length but head rotated $90^\circ/180^\circ$.	M
	Null destination	Head on empty table or off-frame.	E
	Detached tail	Shaft plausible but tail not anchored on the circled object.	H
	Spec mismatch	Arrow present on a circle-only task, or missing on a place task.	H
Cross-modal (channels disagree; v_X)	Circle–arrow mismatch	Circle object A while the arrow originates from object B.	H
	Caption–sketch mismatch	Correct sketch, but caption swapped from another demo/task.	H
	Predicate mismatch	Caption says “open the cabinet” while the sketch is a place-style circle+arrow.	M
Language (caption / “what”; v_L)	Wrong target noun	Caption names the wrong object.	M
	Wrong destination noun	Caption names the wrong destination.	E
	Wrong verb	Caption verb wrong (pick vs. push vs. open).	M
	Under-determined	“Put this there”: valid with a good sketch, invalid with a bad one.	H
Grounding (v_S)	Five configurations	Fully resolved prompt; no sketch; open destination; identical referents; arrow between destinations.	M–H
Physical (deferred)	Occluded target	Circle a target not visible in the frame (needs depth); generator yielded only 18 rows in the pilot.	M

Table 2: DrawVLA-Verify v1 composition by family (train / validation / test).

Family	Train	Val	Test
Referential	6,354	766	834
Directional	7,338	904	958
Cross-modal	1,212	150	158
Language	2,556	308	336
Grounding	5,648	695	655
Positive	7,194	879	1,025
Total	30,302	3,702	3,966

B PILOT RESULTS, TRAINING DETAILS, AND ZERO-SHOT PROTOCOL

Table 3: Pilot results on the 3,966-example v1 test set (single seed; zero-shot rows use one direct prompt; the test set is 74% negative). [†]Shortcut ceiling from raw coordinates only.

Model	Acc	BAcc	Macro-F1	MCC
Majority class (always REJECT)	0.74	0.50	0.43	0.00
Qwen3-VL-2B (zero-shot, open)	0.32	0.54	0.30	0.13
Qwen3-VL-4B (zero-shot, open)	0.46	0.61	0.46	0.22
Claude Sonnet 4.6 (zero-shot, API)	0.67	0.59	0.59	0.18
Claude Haiku 4.5 (zero-shot, API)	0.73	0.50	0.44	0.01
Gemini 3.7 Flash (zero-shot, API)	0.79	0.71	0.71	0.43
SmolVLM-256M + LoRA (binary-only)	0.89	0.83	0.84	0.69
SmolVLM-500M + LoRA (binary-only)	0.90	0.84	0.86	0.72
Geometry-only HGB [†]	0.93	0.91	0.91	–

Fine-tuning. SmolVLM-256M and -500M with LoRA rank 16, $\alpha = 32$, dropout 0.05, 3 epochs, batch size 64, learning rate 10^{-4} with cosine schedule and 3% warm-up. Decisions are read by forced-choice scoring over the RIGHT/WRONG answer tokens rather than free-form generation. The pilot adapters were trained on an earlier 34,018-row release that contained 14 physical-family negatives; all reported metrics are computed on the identical 3,966-row physical-free test set, and the proposed runs will retrain on the physical-free data.

Zero-shot protocol. Each VLM receives the rendered image and a fixed prompt that explains the circle/arrow/caption semantics, states the three validity conditions (correct, consistent, sufficient), and asks for a single-word RIGHT/WRONG answer; the identical prompt is used for open and closed models.

Pilot error profile. SmolVLM-500M makes 415 test errors: 122 false accepts and 293 false rejects. The weakest slices are arrow-between-destinations (83.1% correct rejection), wrong destination (88.0%), wrong magnitude (89.9%), and near-miss circle (91.7%)—exactly the geometrically subtle modes, which motivates H3.